ASAD GULSHAIR

AI Engineer | Machine Learning Engineer | Data Scientist

Lahore, Pakistan | +92 321 6675424 | asadgulshair6767@gmail.com | linkedin.com/in/asad-gulshair | github.com/asad-gulshair

PROFESSIONAL SUMMARY

Results-driven AI & Machine Learning Engineer with a B.S. in Data Science (2025) and hands-on experience building and deploying production-ready AI systems, computer vision pipelines, and LLM-based applications. Proven track record of delivering scalable ML solutions with 90%+ model accuracy, 40% reduction in manual workflows, and measurable business impact. IEEE-published researcher with expertise spanning deep learning, generative AI, MLOps, and business intelligence analytics.

PROFESSIONAL EXPERIENCE

Data Scientist – Intelligence & Innovation | Dot Republic Media – FZCO | Lahore, Pakistan (On-site) *Jan 2026 – Present*

- Performed end-to-end data cleaning, preprocessing, and statistical analysis on structured and unstructured datasets to support business growth initiatives.
- Designed and deployed Power BI / Tableau dashboards and analytics reports to surface actionable insights for executive decision-making.
- Analyzed YouTube Analytics and API data to identify content performance trends, directly improving content strategy and viewer engagement.
- Built automation workflows using n8n, reducing manual operational overhead and streamlining internal processes.
- Engineered AI subtitle generation pipelines using FFmpeg, audio processing, and LLaMA-based translation APIs for multi-language content.
- Implemented automated translation verification systems, improving multilingual content accuracy across markets.

Artificial Intelligence Engineer | RICI Pakistan Pvt. Ltd. | Lahore, Pakistan (On-site) *July 2025 – Dec 2025*

- Built end-to-end AI-powered industrial inspection systems using computer vision and deep learning for manufacturing quality assurance.
- Trained and optimized YOLOv8 and CNN models for welding and coating defect detection, achieving >90% accuracy on production data.
- Reduced manual inspection time by 40% through fully automated real-time defect detection pipelines.
- Implemented precision dimension measurement systems using OpenCV and ArUco markers for geometric quality validation.
- Applied monocular depth estimation techniques to enhance 2D-to-3D visual measurement accuracy.
- Deployed all models via FastAPI REST APIs for seamless integration with production web and mobile applications.

Key Skills: YOLOv8, CNNs, OpenCV, Computer Vision, Depth Estimation, FastAPI, Model Deployment, Data Augmentation

Artificial Intelligence Intern | Evolution Software | Remote *Jun 2025 – July 2025*

- Worked on applied AI projects spanning model development, evaluation, and API deployment.
- Built a Voice Emotion Recognition system using MFCC features and a CNN–LSTM hybrid architecture, improving classification accuracy from 50% to 80%.
- Developed a Drug Recommendation Engine using XGBoost and LightGBM ensemble models, achieving ~90% prediction accuracy.
- Assisted in model optimization and FastAPI-based deployment for production environments.

KEY PROJECTS

AI-Powered Industrial Inspection System · *YOLOv8 · OpenCV · CNN · Flask · FastAPI*

- Designed an end-to-end computer vision system for automated quality inspection in manufacturing environments.
- Trained object detection models for welding and coating defect identification, achieving >90% accuracy with real-time inference.
- Reduced inspection cycle time by 40% through automation, cutting operational costs and human error.
- Deployed scalable REST APIs using FastAPI for seamless integration with production systems.

GitHub: github.com/asad-gulshair/End-to-End-Complete-Inspection-system-for-Welding-Line

AI-Powered Smart Farming System (IEEE-Published Final Year Project) · *Python · TensorFlow · IoT · FastAPI*

- Developed an AI-driven precision agriculture platform monitoring soil health and recommending crops, irrigation, and fertilizer usage.
- Integrated real-time IoT soil sensor data (moisture, pH, NPK) with live weather forecast APIs for dynamic decision support.
- Trained ML/DL models achieving 95–97% accuracy, resulting in a measured 30% improvement in crop yield outcomes.
- Deployed via FastAPI and integrated an AI chatbot interface for farmer assistance and real-time query handling.

GitHub: github.com/asad-gulshair/Agro-high-AI-powered-Forming-Project

RAG-Based Conversational AI Chatbot · *Sentence Transformers · FAISS · FLAN-T5 · Python*

- Built a production-ready Retrieval-Augmented Generation (RAG) pipeline using Sentence Transformers for dense text embeddings and FAISS for efficient semantic search.
- Implemented document chunking and similarity-based retrieval to ground responses in relevant source material.
- Used FLAN-T5 to generate context-aware, hallucination-resistant answers from retrieved documents.

GitHub: github.com/asad-gulshair/Python-RAG-System-Embeddings-FAISS-T5-Pipeline

Voice Emotion Detection System · *Python · Librosa · TensorFlow · CNN*

- Processed 7,000+ audio samples from the CREMA-D dataset; extracted MFCC, delta, and delta-delta features using Librosa for rich spectral representation.
- Designed and trained CNN-based models for 6-class emotion recognition (happy, sad, angry, fearful, disgusted, neutral), achieving 75%+ accuracy.
- Enhanced model performance through systematic spectral feature extraction optimization and CNN architecture tuning.

GitHub: github.com/asad-gulshair/Voice-Emotion-Detection

TECHNICAL SKILLS

ML & Deep Learning	TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, YOLOv8, CNNs, Transformers, Model Evaluation, Cross-Validation, Data Augmentation
GenAI & NLP	LLMs, RAG, llama3, ChromaDB, Hugging Face, Sentence Transformers, FAISS, FLAN-T5, Sentiment Analysis
Computer Vision	OpenCV, Image Processing, Object Detection, Depth Estimation, Transfer Learning, ArUco Markers
Data Science	Statistical Analysis, Feature Engineering, EDA, Pandas, NumPy, SQL, Power BI, Tableau, ETL Pipelines, Data Visualization
Deployment & MLOps	FastAPI, Flask, REST APIs, Docker, Git, Hyperparameter Tuning (Optuna), CI/CD, n8n Automation
Programming	Python, SQL, JavaScript (basic)

EDUCATION

Bachelor of Science in Data Science *2021 – 2025*

Government College University, Faisalabad | CGPA: 3.35 / 4.00

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Big Data Analytics, Statistics, Artificial Intelligence, Database Management

ACHIEVEMENTS & PUBLICATIONS

IEEE Xplore Publication (2025) — Co-authored peer-reviewed research on AI-Driven Smart Agriculture: An Integrated Approach for Soil Analysis, Irrigation, and Crop-Fertilizer Recommendations.

40 Public GitHub Repositories — Active open-source contributor with projects spanning computer vision, CNN, YOLO, LLM, NLP, GenAI, and data science.